Distributed Consensus in Latency-Constrained Edge Networks

Abstract—*This paper investigates robust synchronization paradigms under strict operational bounds. We present theoretical guarantees and empirical results confirming sustained throughput.*

1. Introduction
Recent advances in asynchronous networking necessitate rigorous verification protocols. Edge environments operate under intermittent connectivity and bounded computational budgets.

2. Problem Formulation
Let G = (V, E) denote the topology graph where V represents edge brokers and E represents high-latency transport links.

3. Methodology
We evaluate partitioned consensus using localized heartbeat synchronizations. Packet loss is mitigated through proactive state redundancy.

4. Experimental Evaluation
Our benchmark demonstrates forty percent reduction in round-trip reconciliation overhead across varied network topologies.